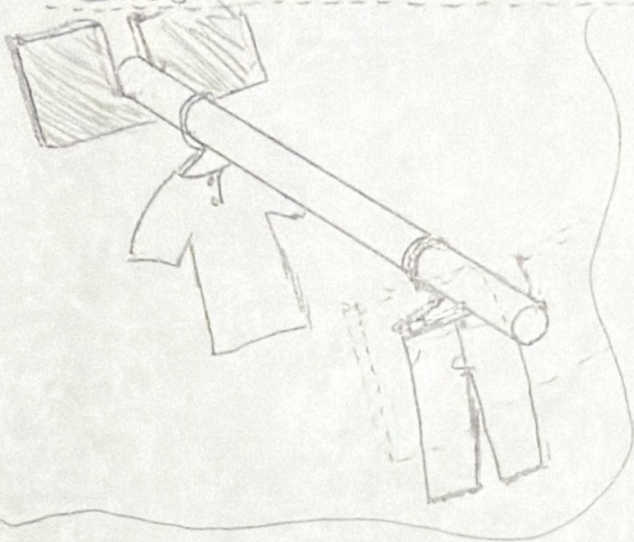
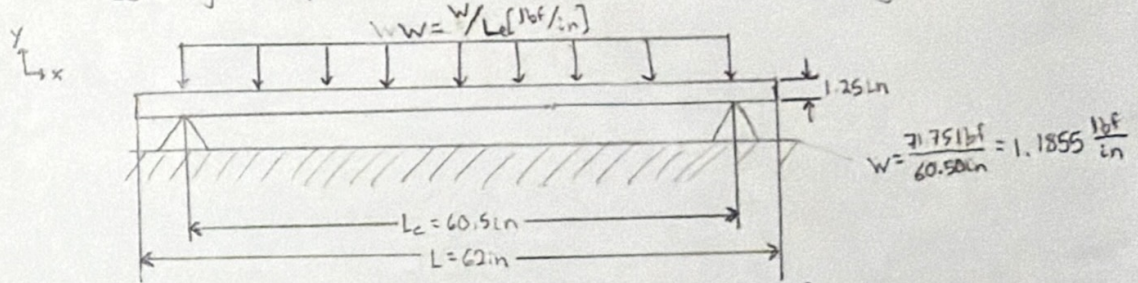


Closet Rod - Preliminary Analysis



Ends modeled as simple supports for circular beam.
Clothing applies total approximate load $W = 71.75 \text{ lbf}$
Rod has diameter $D = 1.25 \text{ in}$ and length $L = 62.00 \text{ in}$
 $\sim 1.5 \text{ in}$ of rod length sits in end-bearings; Effective length $L_e = 60.50 \text{ in}$.
Loading evenly distributed across full effective length.



$$M_{\max} = \frac{wL_e^2}{8} = \frac{WL_e}{8} = \frac{(71.75 \text{ lbf})(60.5 \text{ in})}{8} = 542.609 \text{ lbf}\cdot\text{in}$$

$$I = \frac{\pi}{64} D^4 = \frac{\pi}{64} (1.25 \text{ in})^4 = 0.1198422 \text{ in}^4 \quad C = R = \frac{D}{2} = \frac{1.25 \text{ in}}{2} = 0.625 \text{ in}$$

$$\sigma_{\text{bending, max}} = \frac{M_{\max} C}{I} = \frac{(542.609 \text{ lbf}\cdot\text{in})(0.625 \text{ in})}{0.1198422 \text{ in}^4} = 2829.81 \text{ psi} = 2.83 \text{ ksi}$$

$$FS = n = \frac{S_{\text{max}}}{\sigma_{\text{bending, max}}} = \frac{14080 \text{ psi}}{2829.81 \text{ psi}} = 4.98$$

$$\delta_{\max} = \frac{5wL_e^4}{384EI} = \frac{5WL_e^3}{384EI} = \frac{5(71.75 \text{ lbf})(60.5 \text{ in})^3}{384(1.781 \times 10^6 \text{ psi})(0.1198422 \text{ in}^4)} = 0.9693 \text{ in}$$

$$\delta_{\text{allow}} = \frac{L_e}{240} = \frac{60.5 \text{ in}}{240} = 0.252 \text{ in}$$

$$W_{\max} = \frac{8\delta_{\max} I}{CL_e} = \frac{8S_{\max} I}{CL_e} = \frac{8(14080 \text{ psi})(0.1198422 \text{ in}^4)}{(0.625 \text{ in})(60.5 \text{ in})} = 357.00 \text{ lbf}$$

$$\delta \propto W \rightarrow W_{\text{allow}} = W \left(\frac{\delta_{\text{allow}}}{\delta_{\max}} \right) = (71.75 \text{ lbf}) \left(\frac{0.252 \text{ in}}{0.9693 \text{ in}} \right) = 18.7 \text{ lbf}$$

$$\text{Creep estimate: } \delta_{\text{creep}} \approx 1.5 \delta_{\max} = 1.5(0.9693 \text{ in}) = 1.45 \text{ in}$$

Material: Teakwood

Properties:

- $E = 1.781 \times 10^6 \text{ psi}$ for stiffness/deflection
- Modulus of Rupture, $S_{\max} = 14080 \text{ psi}$ for approximate bending stress failure
- Density, $\rho = 0.0237 \text{ lb/in}^3$
- Poisson's Ratio $\nu = 0.325$

Calculation Description:

- Max bending stress assuming teakwood's modulus of rupture approximately describes bending strength
 - To identify current stress of my clothing
 - $FS(n)$ describes how close current load is to rod failure
- Max deflection (δ) of rod, then common serviceability allowable deflection (before sag)
 - Max load, $W_{\max}$, before rupture occurs
 - Then load before sag
- Approximate long-term deflection due to creep.